LAB 17

Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution

Item	Detail
Topic	Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis
Pipeline stage	Event-driven automation
TSC reference	A7, K5, K7
Duration	20 min
Simulator page	http://localhost:8000/#/integrations
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Some Lion City Capital traders design signals in TradingView. The desk wants their alerts to place paper orders automatically - but only through the pipeline: a gateway that checks a shared secret, the allow-list, a maximum size and duplicate alerts, asks the pipeline's own signal, runs the risk checks and then hands the order to the order manager, logging every request.

Objective

Send TradingView alerts to the app's webhook and see them pass validation, signal confirmation and risk into paper orders.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 17. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 17: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Copy the alert template

Integrations > TradingView webhook: copy the URL and the JSON alert message (reveal the secret only on your own machine).

Open: http://localhost:8000/#/integrations (Integrations > TradingView webhook)

Check: You know which placeholders TradingView fills in.

Step 2. Run the self-test

Press 'Run self-test (7 simulated alerts)'.

Open: http://localhost:8000/#/integrations

Check: 7/7: valid buy and sell (200), wrong secret (401), symbol not allowed and qty too large (422), malformed (400), retry ignored (200 duplicate).

Step 3. Real alerts (paid TradingView plan)

Optional - trainer demonstration: with a paid TradingView plan, an alert on the exported Pine strategy can post to the simulator's webhook URL once the trainer has made the simulator reachable from the internet (a public tunnel URL). Watch the trainer's alert pass validation, signal confirmation and risk into a paper order.

Open: http://localhost:8000/#/integrations

Check: The alert appears in the audit log and its order in Paper Trading > Orders (source webhook).

Expected in the app: 7 simulated alerts handled as expected and the audit log.

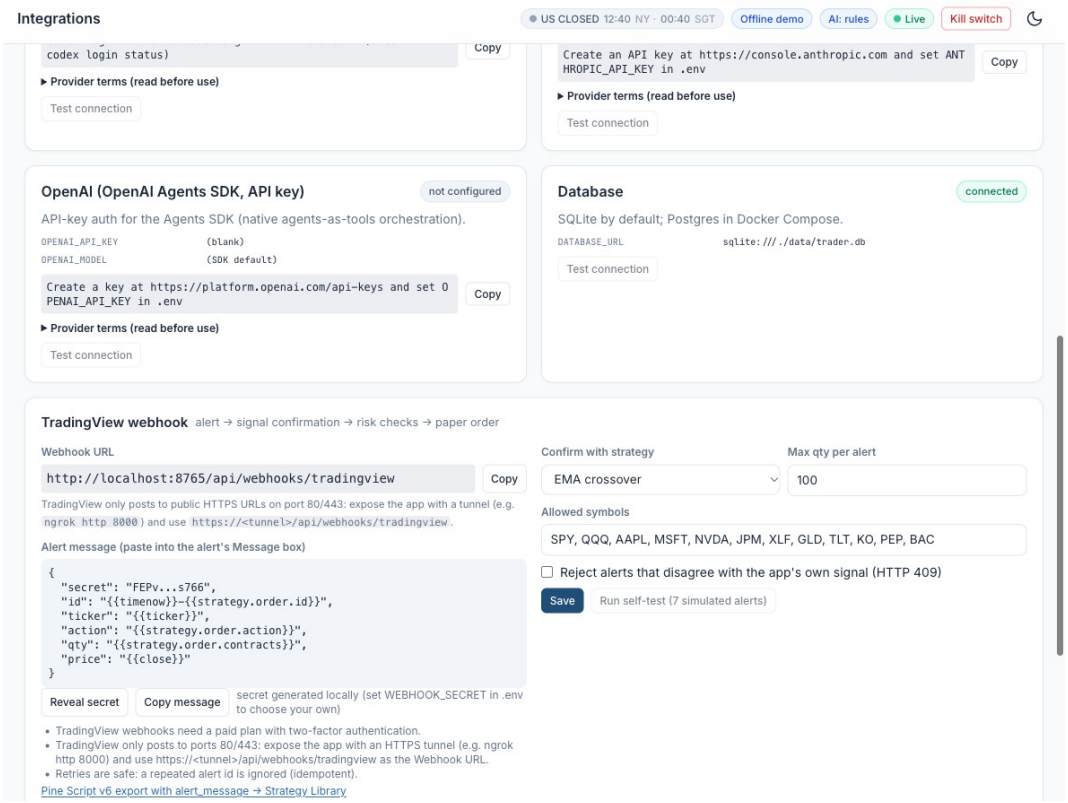


Figure: Lab 17 in the Trading Simulator.

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Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs_lowcode/Prompt_Library.html and .pdf.

A webhook that turns TradingView alerts into paper orders - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: write a small FastAPI app with a /webhook endpoint that receives a TradingView alert JSON {secret, ticker, action, price}, rejects a wrong secret, validates the fields, confirms the signal with a 20-day SMA check, caps the position size, and logs the paper order it would send. Test it inside Colab with FastAPI's TestClient by posting 7 sample alerts (valid buy, valid sell, wrong secret, missing field, unknown ticker, duplicate, oversized) and print each result.

Open in Claude | **Open in ChatGPT** | **Open Google Colab**

Debrief

A webhook lets a charting platform trigger execution on a trading platform (K5), but the gateway code (A7) is where the controls live: authentication, allow-lists, size limits, duplicate protection, the pipeline's own signal, the risk layer and an audit log. Each automated order still pays spread and slippage (K7), so alert frequency matters.

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